

SVENNER FYR, Norway

WMO: 014780

Lat: 58.969N

Lon: 10.148E

Elev: 49

StdP: 14.67

Time Zone: 1.00 (EUC)

Period: 10-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	14.3	17.5	N/A	N/A	N/A	N/A	N/A	N/A	38.3	38.5	36.3	39.6	13.0	10	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.9	73.2	N/A	70.2	N/A	68.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	8.5	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 37.8	33.9	30.7	DB	14.7	76.6	4.8	3.8	11.3	79.3	8.5	81.5	5.8	83.6	2.3	86.3	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.2	32.6	32.3	37.6	43.4	52.3	58.8	63.5	62.0	56.9	48.6	41.0	36.1	(6)	
	DBStd	12.29	6.84	6.85	5.39	4.87	6.39	3.68	3.98	2.85	3.61	4.92	6.51	8.32	(7)	
	HDD50	2460	538	495	386	209	44	0	0	0	2	85	272	429	(8)	
	HDD65	6548	1003	915	850	649	397	191	78	101	242	508	720	894	(9)	
	CDD50	1434	0	0	1	10	116	264	418	372	210	41	2	0	(10)	
	CDD65	47	0	0	0	0	4	4	31	8	0	0	0	0	(11)	
	CDH74	62	0	0	0	0	7	4	51	0	0	0	0	0	(12)	
	CDH80	1	0	0	0	0	0	0	1	0	0	0	0	0	(13)	
(14) Wind	WSAvg	14.4	16.4	15.8	14.2	12.3	12.2	12.7	12.3	13.5	15.2	15.9	16.2	16.6	(14)	
(15) Precipitation	PrecAvg	40.2	3.7	2.7	2.3	2.3	2.5	3.3	3.3	4.0	3.8	4.4	4.2	3.8	(15)	
	PrecMax	55.4	7.5	6.7	5.5	4.5	4.1	6.5	7.2	6.6	7.6	9.8	11.1	7.2	(16)	
	PrecMin	31.1	0.8	1.3	0.4	0.7	0.5	0.5	0.4	1.0	1.6	0.9	1.6	0.9	(17)	
	PrecStd	5.9	1.6	1.3	1.2	1.1	1.1	1.3	1.6	1.4	1.7	2.0	2.0	1.6	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.9	46.0	55.3	61.2	74.1	72.6	78.7	72.3	65.7	59.2	53.9	50.3	(19)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(20)	
	2%	DB	44.6	42.9	48.4	56.9	68.7	68.8	75.0	69.8	64.1	57.9	52.0	48.6	(21)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(22)	
	5%	DB	43.3	41.3	45.4	53.8	64.8	66.4	72.0	68.2	63.0	56.9	50.8	47.4	(23)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(24)	
	10%	DB	41.7	40.2	43.5	50.8	61.6	64.3	69.4	66.7	62.1	55.7	49.4	45.9	(25)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(27)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(28)	
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(29)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(30)	
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(31)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(32)	
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(33)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.7	5.4	7.0	7.9	7.9	7.1	6.9	6.6	6.5	6.3	5.7	6.2	(35)	
		MCDBR	7.0	6.5	10.2	12.5	12.2	10.0	10.4	8.3	6.9	5.8	6.2	5.7	(36)	
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(37)	
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(38)	
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.288	0.323	0.341	0.380	0.373	0.359	0.381	0.378	0.349	0.337	0.304	0.264	(40)	
	taud		2.244	2.323	2.369	2.317	2.414	2.477	2.436	2.438	2.505	2.432	2.292	2.172	(41)	
	Ebn at Noon		167	221	253	261	272	278	268	260	251	214	159	132	(42)	
	Edh at Noon		14	22	27	34	33	31	32	30	24	19	13	10	(43)	
(44) All-Sky Solar Radiation	RadAvg		103	290	762	1240	1637	1865	1681	1314	866	390	136	70	(44)	
	RadStd		20	56	74	170	167	124	186	87	98	65	33	8	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (53 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page